

Carlos J. Soto

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EMPLOYMENT EXPERIENCE

Assistant Professor September 2023 - Present
University of Massachusetts Amherst

Bruce Lindsay Visiting Assistant Research Professor August 2020 - August 2023
Pennsylvania State University

Research position exploring both the theoretical and methodological connections between differential privacy and the geometry of the space in which the data live. Implemented differentially private models on manifolds and evaluated their performance as compared to state of the art techniques. This position also has a teaching component of six credits (two courses) per academic year. Under the supervision of Matthew Reimherr and Aleksandra Slavković.

VISITING POSITIONS

Visiting Researcher Fall 2025
Research Group for Geometric Data Analysis and Processing, Zuse Institute Berlin

Visiting Researcher Fall 2025
Research Group for Statistics, Humboldt University Berlin

EDUCATION

PhD Biostatistics Fall 2017 – July 2020
Florida State University Tallahassee, Florida

- Cumulative GPA: 4.0
- Dissertation “Structural Data Analysis in Bioinformatics: With a Focus on Chromosomes and Proteins”
- Advised by Anuj Srivastava

MS Biostatistics Fall 2015 – Fall 2017
Florida State University Tallahassee, Florida

- Cumulative GPA: 4.0

MS Mathematics Fall 2013 – Spring 2015
University of Wisconsin–Milwaukee Milwaukee, Wisconsin

- Cumulative GPA: 3.618

BA Mathematics Fall 2007 – Spring 2011
Ripon College Ripon, Wisconsin

- Cumulative GPA: 3.42, Graduated Cum Laude

PUBLICATIONS

1. Carlos Soto, Cheng Wang, Zipan Huang, and Xiaoyu Chen. Intrinsic riemannian cross-covariance for manifold-valued random objects. In *The Fortieth Annual Conference on Neural Information Processing Systems*, 2026
2. Aditya Kulkarni and Carlos Soto. Differentially private geodesic and linear regression. In *Proceedings of the 43rd International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2026
3. Cheng Wang and Carlos J Soto. Efficient recursive fr\'echet mean estimation. In *The 42nd Conference on Uncertainty in Artificial Intelligence*, 2026
4. Carlos J Soto, Matthew Reimherr, Aleksandra Slavkovic, and Mark Shriver. Gaussian differentially private human faces under a face radial curve representation. In *The Thirteenth International Conference on Learning Representations*, 2025
5. Carlos Soto, Karthik Bharath, Matthew Reimherr, and Aleksandra Slavkovic. Shape and structure preserving differential privacy. *Advances in Neural Information Processing Systems*, 2022
6. Carlos Soto, Darshan Bryner, Audrey Dalgarno, Nicola Neretti, and Anuj Srivastava. Tadbay: A bayesian construction of topologically associated domains. In *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 233–240. IEEE, 2022
7. Carlos Soto, Darshan Bryner, Nicola Neretti, and Anuj Srivastava. Toward a three-dimensional chromosome shape alphabet. *Journal of Computational Biology*, pages 601–618, 2021
8. Carlos J Soto, Peiyao A Zhao, Kyle N Klein, David M Gilbert, and Anuj Srivastava. Statistical comparisons of chromosomal shape populations. In *2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI)*, pages 788–791. IEEE, 2021
9. Matthew Reimherr, Karthik Bharath, and Carlos Soto. Differential privacy over riemannian manifolds. *Advances in Neural Information Processing Systems*, 34, 2021
10. Carlos Soto, Audrey Dalgarno, Darshan Bryner, Benjamin McLaughlin, Nicola Neretti, and Anuj Srivastava. Representation of chromosome conformations using a shape alphabet across modeling methods. In *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 151–156, 2021
11. Jose Cordova, Carlos Soto, Mostafa Gilanifar, Yuxun Zhou, Anuj Srivastava, and Reza Arghandeh. Shape preserving incremental learning for power systems fault detection. *IEEE control systems letters*, 3(1):85–90, 2018

In Review

1. Aditya Kulkarni, Davide Sartor, Carlos J Soto, and Nathan Wycoff. How low can you go: Fine-tuning llms with bayesian optimization (llmbo)

In Progress

1. Carlos Soto. Rao differential privacy. *arXiv preprint arXiv:2508.17135*, 2025
2. Carlos Soto, et al. “Differential Privacy over Lie Groups.”
3. Cheng Wang and Carlos J Soto. “Shape-in-shape-out regression.”

INVITED TALKS AND PRESENTATIONS

‡ ICML (International Conference on Machine Learning)

July 6-11, 2026

Differentially Private Geodesic Regression

Seoul

* New England Statistics Symposium (NESS)

May 28-29, 2026

Rao Differential Privacy

Storrs, CT

* CFE-CMStatistics	Dec 13-15, 2025
<i>Differentially Private Geodesic Regression</i>	London, UK
‡ ICLR (International Conference on Learning Representations)	April 25-28, 2025
<i>Gaussian Differential Privacy for Human Faces</i>	Singapore
* Joint Statistical Meetings (JSM)	August 7, 2025
<i>Differentially Private Geodesic Regression</i>	Nashville, TN
* National University Singapore	April 24, 2025
<i>Differential Privacy on Shape Manifolds</i>	Singapore
* AMS Sectional Meeting	April 13, 2025
<i>Representation of Chromosome Conformations Using a Shape Alphabet Across Modeling</i>	Hartford, CT
* CFE-CMStatistics	Dec 14-16, 2024
<i>Gaussian Differential Privacy for Human Faces</i>	London, UK
* New England Statistics Symposium (NESS)	May 23-24, 2024
<i>Representation of Chromosome Conformations Using a Shape Alphabet Across Modeling</i>	Storrs, CT
* IMS International Conference of Statistics and Data Science (ICSDS)	Dec, 2023
<i>Shape Preserving Differential Privacy</i>	Lisbon, Portugal
* Joint Statistical Meetings (JSM)	Aug, 2023
<i>Differential Privacy Over Riemannian Manifolds</i>	Toronto, Canada
* National Agricultural Statistics Service (USDA NASS)	July, 2023
<i>Differential Privacy Over Riemannian Manifolds</i>	Virtual
‡ NeurIPS (Neural Information Processing Systems)	Dec 2022
<i>Shape And Structure Preserving Differential Privacy</i>	New Orleans, LA
* The 35th New England Statistics Symposium	May 22-25, 2022
<i>Geometry-driven Statistics: Differential Privacy on Manifolds</i>	Storrs (UConn), CT, USA
Workshop on Differential Privacy and Statistical Data Analysis	July 25-29, 2022
<i>Intrinsic Differential Privacy</i>	Fields Institute (Toronto), ON, CAN
Computational and Methodological Statistics	Dec 18-20, 2021
<i>Recent advances in differential privacy: Differential privacy over Riemannian manifolds</i>	London, UK
Stochastic Modeling and Computational Statistics (SMAC)	Apr 21, 2023
<i>Shape and Structure preserving Differential Privacy</i>	University Park, PA
* Symposium on Data Science and Statistics (SDSS)	Jun 7-10, 2022
<i>Differential Privacy on Manifolds</i>	Pittsburgh, PA
Joint Statistical Meetings(JSM)	Aug 8, 2022
<i>Shape and Structure Preserving Differential Privacy on Manifolds</i>	Washington D.C.
* IEEE-BIBM(International Conference on Bioinformatics and Biomedicine)	Dec 9-12, 2021
<i>Representation of Chromosome Conformations Using a Shape Alphabet Across Modeling Methods</i>	Virtual
Joint Math Meetings (JMM)	April 6, 2022
<i>Differential Privacy Over Riemannian Manifolds</i>	Virtual
IEEE-ISBI (International Symposium of Biomedical Imaging)	April 13-16, 2021
<i>Statistical Comparisons of Chromosomal Shape Populations</i>	Virtual
‡ NeurIPS (Neural Information Processing Systems)	Dec 2021
<i>Differential Privacy Over Manifolds</i>	Virtual
Stochastic Modeling and Computational Statistics (SMAC)	Dec 3, 2021
<i>Differential Privacy Over Riemannian Manifolds</i>	University Park, PA
Joint Statistical Meetings(JSM)	August 5, 2020

TEACHING EXPERIENCE

Instructor Spring 2026
University of Massachusetts Amherst Amherst, MA
Full instructor for STAT 315: Introduction to Statistics. The first of a sequence; covers probability theory.

Instructor Spring 2025
University of Massachusetts Amherst Amherst, MA
Full instructor for STAT 690S: Statistical Shape Analysis.

Instructor Fall 2024
University of Massachusetts Amherst Amherst, MA
Full instructor for STAT 535: Statistical Computing. Introductory statistical computing in R.

Instructor Spring 2024
University of Massachusetts Amherst Amherst, MA
Full instructor for STAT 535: Statistical Computing. Introductory statistical computing in R.

Instructor Fall 2023
University of Massachusetts Amherst Amherst, MA
Full instructor for STAT 515: Introduction to Statistics. The first of a sequence; covers probability theory.

Instructor Spring 2021, Fall 2021, Fall 2022
Pennsylvania State University University Park, PA
Full instructor for STAT 380: Data Science Through Statistical Reasoning and Computation, responsible the entire course except grading.

Instructor Fall 2016 – Spring 2020
Florida State University Tallahassee, Florida
Full instructor for STA 2171: Statistics for Biology, responsible for lecturing and creating all coursework including worksheets and exams.

Teaching Assistant Fall 2015 – Spring 2016
Florida State University Tallahassee, Florida
Teaching Assistant for CGS 2518: Spreadsheets for Business, responsible for assisting students with assignments as well as grading assignments and exams.

Instructor Fall 2013 – Spring 2015
University of Wisconsin–Milwaukee Milwaukee, Wisconsin
Full instructor for MATH 098, MATH 108, and MATH 105, responsible for lecturing and creating all coursework including worksheets and exams.

Math and Statistics Tutor and Assistant Fall 2009 – Spring 2011
Ripon College Ripon, Wisconsin
Assisted fellow undergraduate students in math and statistics courses including multivariate calculus, linear algebra, and introductory statistics.

SERVICE

- Organizer of the 2026 Institute of Mathematical Statistics (IMS) New Researchers Conference (NRC).
- Steering Committee member (2024), chair-elect (2025), and chair (2026) for the ASA Privacy and Confidentiality Interest Group.
- Organizer (UMass) of the ASA Five College Datafest 2024, 2025.

- Reviewer for *International Conference on Learning Representations (ICLR)*, *Annals of Applied Statistics*, *Institute of Electrical and Electronics Engineers/Association for Computing Machinery (IEEE/ACM)*, *Institute of Electrical and Electronics Engineers/International Symposium on Biomedical Imaging (IEEE/ISBI)*, *Computational and Structural Biotechnology Journal*, *Neural Information Processing Systems (NeurIPS)*, *International Conference on Machine Learning (ICML)*, *Journal of the American Statistical Association (JASA)*, *Journal of Multivariate Analysis (JMVA)*, *Journal on Uncertainty Quantification (JUQ)* and *Journal of the Royal Statistical Society: Series C (JRSSC)*.
- Member of the Professional Development Committee of the ASA JEDI (American Statistical Association - Justice Equity Diversity and Inclusion) 2023-2024.
- Member of the Penn State Statistics Department Climate and Diversity Committee 2021-2023.
- Organizer of Penn State's Statistics Department Differential Privacy group Fall 2021.
- ASA Student Chapter President - Florida State University 2018.

MEMBERSHIPS

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- Institute of Electrical and Electronics Engineers (IEEE)
- Mathematical Association of America (MAA)
- American Mathematical Society (AMS)
- New England Statistical Society (NESS)

SKILLS

Proficient	MATLAB, R, Python, L ^A T _E X
Familiar	C++, SAS, and SQL
Languages	English (native), Spanish (fluent)

AWARDS

- IMS ICSDS 2024 Junior Researcher Travel Support